

8. Demonstrate CNN algorithm by taking suitable data for a simple application.

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import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.datasets import mnist

# Load dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Preprocessing
X_train = X_train / 255.0
X_test = X_test / 255.0

# Reshape to CNN format (28x28x1)
X_train = X_train.reshape(-1, 28, 28, 1)
X_test = X_test.reshape(-1, 28, 28, 1)

# Build CNN model
model = models.Sequential()

# Convolution + Pooling Layer 1
model.add(layers.Conv2D(32, (3,3), activation='relu',
input_shape=(28,28,1)))
model.add(layers.MaxPooling2D((2,2)))

# Convolution + Pooling Layer 2
model.add(layers.Conv2D(64, (3,3), activation='relu'))
model.add(layers.MaxPooling2D((2,2)))

# Flatten
model.add(layers.Flatten())

# Fully Connected Layer
model.add(layers.Dense(64, activation='relu'))

# Output Layer
model.add(layers.Dense(10, activation='softmax'))

# Compile model
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Train model
model.fit(X_train, y_train, epochs=5)
# Evaluate
test_loss, test_acc = model.evaluate(X_test, y_test)

print("Test Accuracy:", test_acc)

```